

Department of Mathematics

Abstract Algebra / Group Theory

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Topic: Center of Dihedral Groups $Z(D_{2n})$

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Problem Statement (Chapter 2, Problem 7)

Let $n \in \mathbb{Z}$ with $n \geq 3$. Prove the following:

- (a) $Z(D_{2n}) = 1$ if n is odd.
- (b) $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$.

Complete and Rigorous Proof

1. Preliminaries and Presentation of D_{2n}

The dihedral group D_{2n} of order $2n$ is the symmetry group of a regular n -gon. It is generated by a rotation r of order n and a reflection s of order 2, with the standard presentation:

$$D_{2n} = \langle r, s \mid r^n = 1, s^2 = 1, sr = r^{-1}s \rangle.$$

From the relation $sr = r^{-1}s$, an easy induction on $i \geq 0$ yields the general commutation formula:

$$sr^i = r^{-i}s \quad \text{for all } i \in \mathbb{Z}. \quad (1)$$

Every element $z \in D_{2n}$ can be uniquely expressed in canonical form as:

$$z = r^i s^j \quad \text{where } 0 \leq i < n \text{ and } j \in \{0, 1\}.$$

Thus, the $2n$ elements of D_{2n} partition into:

- **Rotations:** r^i for $0 \leq i < n$ (the cyclic subgroup $\langle r \rangle \cong \mathbb{Z}_n$),
- **Reflections:** $r^i s$ for $0 \leq i < n$ (elements of order 2).

2. Characterization of the Center $Z(D_{2n})$

The center of a group G is the subgroup of elements commuting with every element in G :

$$Z(D_{2n}) = \{z \in D_{2n} \mid zg = gz \quad \forall g \in D_{2n}\}.$$

Since $D_{2n} = \langle r, s \rangle$, an element $z \in D_{2n}$ belongs to $Z(D_{2n})$ if and only if it commutes with both generators r and s :

$$z \in Z(D_{2n}) \iff zr = rz \quad \text{and} \quad zs = sz.$$

3. Exhaustive Case Analysis

Case I: Elements of the form $z = r^i s$ (Reflections)

Let $z = r^i s$ for some $0 \leq i < n$. We test whether z commutes with the generator r :

$$zr = (r^i s)r = r^i(sr) = r^i(r^{-1}s) = r^{i-1}s,$$

$$rz = r(r^i s) = r^{i+1}s.$$

For $z \in Z(D_{2n})$, we require $zr = rz$:

$$r^{i-1}s = r^{i+1}s \implies r^{i-1} = r^{i+1} \implies r^2 = 1 \implies n \mid 2.$$

However, by hypothesis, $n \geq 3$, which strictly precludes $n \mid 2$. Thus $r^2 \neq 1$, which implies:

$$zr \neq rz \quad \text{for all } 0 \leq i < n.$$

Conclusion for Case I: No reflection $r^i s$ belongs to the center $Z(D_{2n})$.

Case II: Elements of the form $z = r^i$ (Rotations)

Let $z = r^i$ for some $0 \leq i < n$.

1. Commutation with r :

$$zr = r^i r = r^{i+1} = r r^i = rz.$$

Thus, every rotation r^i commutes with r .

2. Commutation with s :

$$zs = r^i s, \quad sz = sr^i = r^{-i} s \quad (\text{using Eq. (1)}).$$

For $z \in Z(D_{2n})$, we require $zs = sz$:

$$r^i s = r^{-i} s \iff r^i = r^{-i} \iff r^{2i} = 1 \iff 2i \equiv 0 \pmod{n} \iff n \mid 2i.$$

Since $0 \leq i < n$, we have $0 \leq 2i < 2n$. Therefore, the only possible multiples of n for $2i$ in this range are $2i = 0$ or $2i = n$:

- **If $2i = 0$:** This gives $i = 0$, corresponding to the identity element $z = r^0 = 1 \in Z(D_{2n})$.
- **If $2i = n$:**
 - **Part (a) — n is odd:** If n is odd, then $\gcd(2, n) = 1$, so $n \mid 2i \implies n \mid i$. Since $0 \leq i < n$, the only solution is $i = 0$. Hence, no non-identity rotation commutes with s .

$$Z(D_{2n}) = \{1\} = 1 \quad \text{if } n \text{ is odd.}$$

- **Part (b) — n is even ($n = 2k$ for an integer $k \geq 2$):** Substituting $n = 2k$ into $2i = n$ gives:

$$2i = 2k \implies i = k.$$

Since $1 \leq k < 2k = n$, this yields a unique non-identity element $z = r^k$. Geometrically, r^k represents a half-turn (180° rotation) which inverts all vectors in the plane and thus commutes with every reflection and rotation.

$$Z(D_{2n}) = \{1, r^k\} \quad \text{if } n = 2k.$$

4. Summary of Results

Combining both cases:

- (a) If n is odd, $Z(D_{2n}) = 1$.
- (b) If $n = 2k$, $Z(D_{2n}) = \{1, r^k\}$.

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